

AC and Transient Analysis

A Formula Reference for Linear Circuits — Inductors, Capacitors, Impedance, Charge, Flux, Voltage & Current

This reference collects the constitutive relations of passive elements, the governing equations of first- and second-order transients, and the phasor/impedance description of sinusoidal steady state. Throughout, the passive sign convention is used: current i enters the terminal at which the voltage v is taken to be positive. The imaginary unit is written j so as not to clash with current, and $\omega = 2\pi f$ denotes angular frequency.

1 Constitutive Relations of Passive Elements

The three ideal two-terminal elements are defined by how voltage and current relate at their terminals.

1.1 Resistor

A resistor obeys Ohm's law instantaneously; it stores no energy.

$$v(t) = R i(t), \quad i(t) = \frac{v(t)}{R}, \quad p(t) = v i = i^2 R = \frac{v^2}{R}.$$

1.2 Capacitor — charge and current

A capacitor stores energy in its electric field. Its defining variable is the stored charge q , proportional to the terminal voltage through the capacitance C :

$$q(t) = C v(t), \quad i(t) = \frac{dq}{dt} = C \frac{dv}{dt}, \quad v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau.$$

With a known initial voltage $v(t_0)$,

$$v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(\tau) d\tau.$$

The voltage across a capacitor cannot change instantaneously for a finite current, so v_C is a continuous state variable.

1.3 Inductor — flux and voltage

An inductor stores energy in its magnetic field. Its defining variable is the flux linkage λ , proportional to the current through the inductance L :

$$\lambda(t) = L i(t), \quad v(t) = \frac{d\lambda}{dt} = L \frac{di}{dt}, \quad i(t) = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau.$$

With a known initial current $i(t_0)$,

$$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(\tau) d\tau.$$

The current through an inductor cannot change instantaneously for a finite voltage, so i_L is a continuous state variable.

1.4 Stored energy

$$W_C(t) = \frac{1}{2} C v^2 = \frac{q^2}{2C}, \quad W_L(t) = \frac{1}{2} L i^2 = \frac{\lambda^2}{2L}.$$

	Resistor	Capacitor	Inductor
v - i law	$v = Ri$	$i = C dv/dt$	$v = L di/dt$
State variable	—	v_C (charge $q = Cv$)	i_L (flux $\lambda = Li$)
Stored energy	0	$\frac{1}{2} C v^2$	$\frac{1}{2} L i^2$
Cannot jump	—	voltage	current

2 First-Order Transients

A circuit with a single energy-storage element reduces, via Thévenin reduction of the resistive part, to a first-order linear ODE. Let $x(t)$ be the state variable (v_C or i_L).

2.1 General first-order form

$$\frac{dx}{dt} + \frac{1}{\tau} x = \frac{1}{\tau} x_\infty, \quad x(t) = x_\infty + [x(0^+) - x_\infty] e^{-t/\tau}.$$

Here $x(0^+)$ is the value just after the switching instant, x_∞ is the final (steady-state) value, and τ is the time constant.

2.2 Time constants

$$\tau_{RC} = R_{th} C, \quad \tau_{RL} = \frac{L}{R_{th}}.$$

R_{th} is the Thévenin resistance seen by the storage element with independent sources deactivated.

2.3 RC circuit: charging and discharging

For a capacitor charged through R from a source V_s (initially uncharged):

$$v_C(t) = V_s \left(1 - e^{-t/RC}\right), \quad i(t) = \frac{V_s}{R} e^{-t/RC}.$$

For a capacitor discharging from V_0 through R :

$$v_C(t) = V_0 e^{-t/RC}, \quad i(t) = -\frac{V_0}{R} e^{-t/RC}.$$

2.4 RL circuit: energizing and de-energizing

For an inductor energized through R from V_s (initially zero current):

$$i_L(t) = \frac{V_s}{R} \left(1 - e^{-Rt/L}\right), \quad v_L(t) = V_s e^{-Rt/L}.$$

For an inductor de-energizing from I_0 through R :

$$i_L(t) = I_0 e^{-Rt/L}, \quad v_L(t) = -I_0 R e^{-Rt/L}.$$

After 5τ the response is within 0.7% of its final value and is treated as settled.

3 Second-Order Transients (Series & Parallel RLC)

With both L and C present the natural response satisfies a second-order ODE. Writing the source-free equation for the state variable x :

$$\frac{d^2x}{dt^2} + 2\alpha \frac{dx}{dt} + \omega_0^2 x = 0, \quad s^2 + 2\alpha s + \omega_0^2 = 0, \quad s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}.$$

3.1 Defining parameters

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ (undamped natural freq.)}, \quad \zeta = \frac{\alpha}{\omega_0} \text{ (damping ratio)}, \quad Q = \frac{1}{2\zeta}.$$

Quantity	Series RLC	Parallel RLC
Neper frequency α	$\frac{R}{2L}$	$\frac{1}{2RC}$
Resonant freq. ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor Q	$\frac{1}{R} \sqrt{\frac{L}{C}}$	$R \sqrt{\frac{C}{L}}$

3.2 Damping cases

- **Overdamped** ($\alpha > \omega_0$): $x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$.
- **Critically damped** ($\alpha = \omega_0$): $x(t) = (A_1 + A_2 t) e^{-\alpha t}$.
- **Underdamped** ($\alpha < \omega_0$): $x(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$, with damped frequency $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$.

4 Sinusoidal Steady State and Phasors

A sinusoid $x(t) = X_m \cos(\omega t + \phi)$ is represented by the phasor $\mathbf{X} = X_m e^{j\phi} = X_m \angle \phi$, recovered through

$$x(t) = \text{Re}\{\mathbf{X} e^{j\omega t}\}.$$

Time differentiation maps to multiplication by $j\omega$, and integration to division by $j\omega$:

$$\frac{d}{dt} \longleftrightarrow j\omega, \quad \int dt \longleftrightarrow \frac{1}{j\omega}.$$

4.1 Impedance and admittance

Impedance $Z = V/I$ generalizes resistance to the complex frequency domain; admittance is its reciprocal $Y = 1/Z = G + jB$.

$$Z_R = R, \quad Z_L = j\omega L, \quad Z_C = \frac{1}{j\omega C} = -\frac{j}{\omega C}.$$

In rectangular and polar form,

$$Z = R + jX = |Z| e^{j\theta}, \quad |Z| = \sqrt{R^2 + X^2}, \quad \theta = \arctan \frac{X}{R},$$

where X is the reactance: $X_L = \omega L > 0$ for an inductor and $X_C = -1/(\omega C) < 0$ for a capacitor.

4.2 Series and parallel combination

$$Z_{\text{series}} = \sum_k Z_k, \quad \frac{1}{Z_{\text{parallel}}} = \sum_k \frac{1}{Z_k}, \quad Z_{1\parallel 2} = \frac{Z_1 Z_2}{Z_1 + Z_2}.$$

4.3 Resonance

Series and parallel RLC circuits resonate when the reactive parts cancel, $\omega_0 L = 1/(\omega_0 C)$:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad f_0 = \frac{1}{2\pi\sqrt{LC}}, \quad \text{bandwidth BW} = \frac{\omega_0}{Q}.$$

5 AC Power

For voltage and current phasors $\mathbf{V} = V_m \angle \theta_v$, $\mathbf{I} = I_m \angle \theta_i$, define the phase difference $\theta = \theta_v - \theta_i$ and RMS magnitudes $V_{\text{rms}} = V_m/\sqrt{2}$, $I_{\text{rms}} = I_m/\sqrt{2}$.

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* = P + jQ, \quad |\mathbf{S}| = V_{\text{rms}} I_{\text{rms}}.$$

$$P = V_{\text{rms}} I_{\text{rms}} \cos \theta \quad (\text{real, W}), \quad Q = V_{\text{rms}} I_{\text{rms}} \sin \theta \quad (\text{reactive, var}).$$

The power factor is $\text{pf} = \cos \theta = P/|\mathbf{S}|$ — lagging for inductive loads ($\theta > 0$) and leading for capacitive loads ($\theta < 0$).